

PRACTICAL 0

AIM: INTRODUCTION TO DBMS LAB.

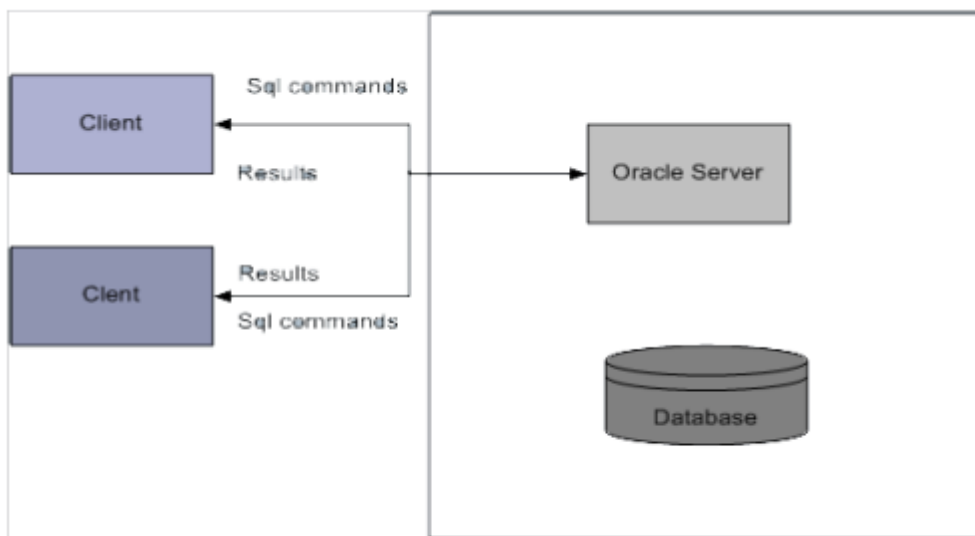
THEORY:

DBMS: A database management system (DBMS) is system software for creating and managing databases. DBMS provides users and programmers with a systematic way to make it possible for end users to create, read, update and delete data in a database. It essentially serves as an interface between the database and end users or application programs, ensuring that data is consistently organized and remains easily accessible.

RDBMS: A relational database management system (RDBMS) is a program that let create, update, and administer a relational database. Most commercial RDBMS's use the Structured Query Language (SQL) to access the database. The leading RDBMS products are Oracle, IBM's DB2 and Microsoft's SQL Server.

Overview of Oracle

Oracle consists of a comprehensive set of application building and end-user products. Oracle applications are accessible on a wide range of platforms and operating system. Oracle provides a flexible Database Management system. Oracle application may run on the same computers as the oracle server. Alternatively application and the tools supporting them may run on system local to the user (the client system), the oracle DBMS runs on another (server system).



Oracle Server

Oracle is the name of a database management system developed by Oracle Corporation. Oracle server manages data in the database. Users access oracle server using SQL commands. So Oracle Server receives SQL commands from users and executes them on the database.

The oracle server comprises a DBMS that controls

- Storage of data in designated database area.
- Retrieval of data for applications, using approximate optimization technique.
- Database security and the tasks permitted for specific users.

- Consistency and protection of data, including archiving task and locking mechanisms.
- Communication and integrity of data when databases are distributed across a network.

Features of Oracle

- Large database support
- Data concurrence
- Industry accepted standards
- Portability
- Enforced Integrity
- Data security
- Support for client/Server environment

SQL

SQL is a powerful query language. The greatest strength of SQL is that it is a standard that most vendors of DBMS software support. Pronounce SQL as sequel. It's the interface between the Kernel and ORACLE utilities and application development tools. SQL is a language that operates on entire groups of data and treats any quantity of information extracted as a single unit. SQL consists of a small number of high level commands that left you query a database to extract and update information stored in a database. The SQL standard is defined by ANSI to which all competing SQL products are supposed to confirm.

SQL is usually provided in two modes – interactive and embedded SQL. Interactive SQL is used to operate directly on a database. The response to any SQL command can be seen almost immediately on the same terminal. Embedded SQL consists of SQL commands used within programs written in some other language like C, ASP, JSP, VB or Pascal.

SQL commands are categorized into the following language statement

➤ **DDL is Data Definition Language statements.**

Some examples:

CREATE - to create objects in the database.

ALTER - alters the structure of the database.

DROP - delete objects from the database.

TRUNCATE - remove all records from a table, including all spaces allocated for the records.

➤ **TCL is Transaction control language**

GRANT - gives user's access privileges to database

REVOKE - withdraw access privileges given with the GRANT command

➤ **DML is Data Manipulation Language statements.**

Some examples:

SELECT - retrieve data from a database.

INSERT - insert data into a table.

UPDATE - updates existing data within a table.

DELETE - deletes all records from a table, the space for the records remain

LOCK TABLE - control concurrency.

➤ **DCL is Data Control Language statements.**

Some examples:

COMMIT - save the changes permanently.

SAVEPOINT - identify a point in a transaction to which you can later roll back

ROLLBACK - restore database to original since the last COMMIT

SET TRANSACTION - Change transaction options like what rollback segment to use.

Data Types used in SQL

Each column contains the datatype and size of the column. The list of available data types in Oracle is shown in table below

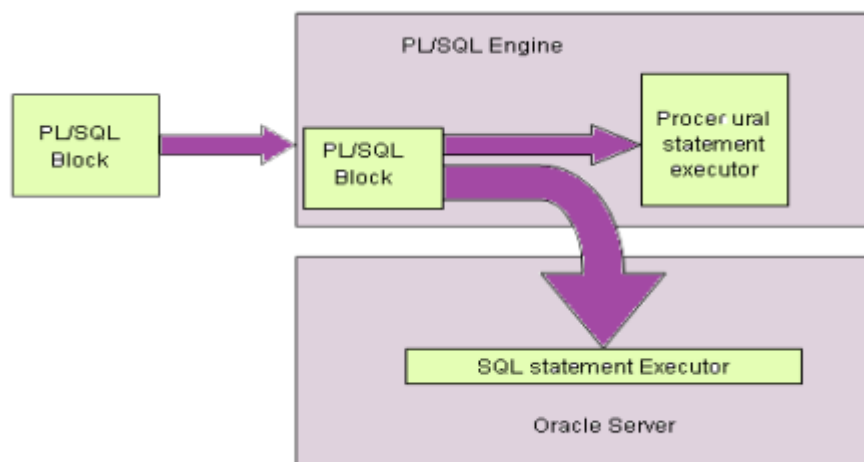
Data type	Description
VARCHAR2(LEN)	Variable length char. Values up to width length. Maximum width is 4000 chars.
CHAR(LEN)	Fixed length char. Values of width length. Default width is 1. Maximum length is 200.
NUMBER	Floating point with precision of 38 significant digits.
NUMBER (Precision, Scale)	Precision represents Maximum significant digits allowed, which may not exceed 38. Scale is the number of decimal places on the right of the decimal point.
DATE	Date values in the range 1-1-4712 B.C to 31-12-4712 AD.
LONG	Variable length char. Values up to 2 GB. Only one LONG col. is allowed per table. You cannot use LONG datatype in functions, WHERE clause and sub queries.
RAW AND LONG RAW	Equivalent to VARCHAR2 and LONG respectively, used for storing digital sound or graphics images.
CLOB, BLOB, NCLOB	Used to store large char. And binary objects. Each can accommodate upto 4 GB.
BFILE	Stores a pointer to an external file.
TIMESTAMP	It is an extension of Date datatype. It stores year, month, day, hour, minute and second values

PL/SQL

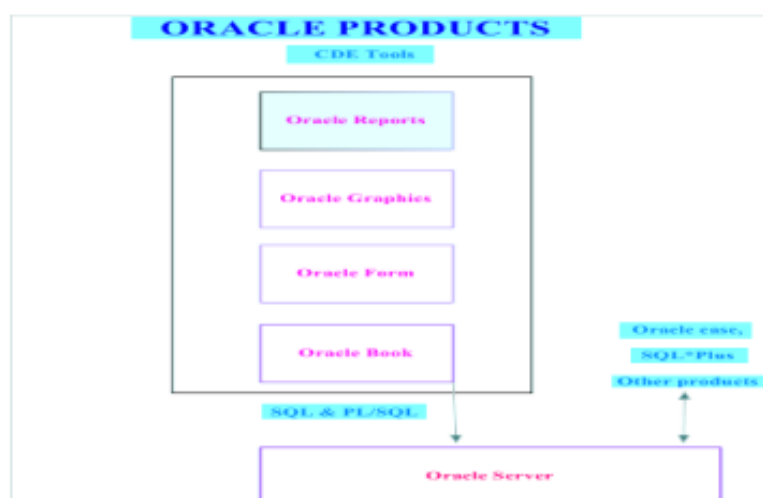
PL/SQL stands for procedural language extension to SQL. PL/SQL allows mixing SQL statements with procedural constructs like IF, looping etc. PL/SQL provides the capability to define and execute PL/SQL blocks such as procedures, functions, triggers and packages. PL/SQL is available only as an "enabling technology" within other software products; it does not exist as a standalone language. You can use PL/SQL in the Oracle relational database, in the Oracle Server, and in client-side application development tools, such as Oracle Forms. Here are some of the ways to use PL/SQL:

- To build stored procedures.
- To create database triggers.
- To implement client-side logic in your Oracle Forms application.
- To link a World Wide Web home page to an Oracle database.

PL/SQL is integrated tightly to the SQL language, yet it adds programming constructs which are not native to this standard relational database language. The programs of PL/SQL looks like traditional 3GL modules, but can be used to include calls to SQL statements, manipulate data through cursors, and take advantage of some of the newest developments in programming languages. PL/SQL supports packages, which allow you to perform object-oriented design. PL/SQL provides a powerful mechanism for trapping and, with exception handlers, resolving errors.



Oracle Products



INSTALLATION PROCEDURE OF ORACLE 10g SOFTWARE:

How to get Oracle 10g :

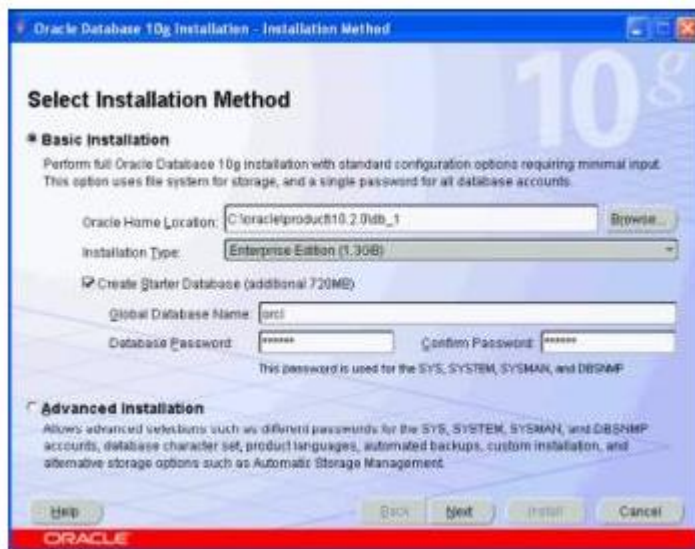
You can download Oracle 10g database from www.oracle.com. You must registered and create an account before you can download the software. The example in this document uses Oracle Database 10g Release 2 (10.2.0.1.0) for Microsoft Windows.

How to uninstall Oracle database software :

1. Uninstall all Oracle components using the Oracle Universal Installer (OUI).
2. Run regedit.exe and delete the **HKEY_LOCAL_MACHINE/SOFTWARE/ORACLE** key. This contains registry entire for all Oracle products.
3. Delete any references to Oracle services left behind in the following part of the registry: **HKEY_LOCAL_MACHINE/ SYSTEM/ CurrentControlsSet/ Services/Ora***. It should be pretty obvious which ones relate to Oracle
4. Reboot your machine.
5. Delete the C: \Oracle directory, or whatever directory is your Oracle_Base.
6. Delete the C:\Program Files \Oracle directory.
7. Empty the contents of your c:\temp directory.
8. Empty your recycle bin.

Installing Oracle 10g database software :

1. Insert Oracle CD , the autorun window opens automatically. If you are installing from network or hard disk, click setup.exe in the installation folder.
2. The Oracle Universal Installer (OUI) will run and display the **Select Installation Method** Window.



3. Choose Basic Installation:

Select this option to quickly install Oracle Database 10g. This method requires minimal user input. It installs the software and optionally creates a general-purpose database based on the information you provide. For **basic installation**, you specify the following:

Oracle Home Location — Enter the directory in which to install the Oracle Database 10g software. You must specify a new Oracle home directory for each new installation of Oracle Database 10g. Use the default value, which is :

c:\oracle\product\10.2.0\db_1

Installation Type — Select Enterprise Edition :

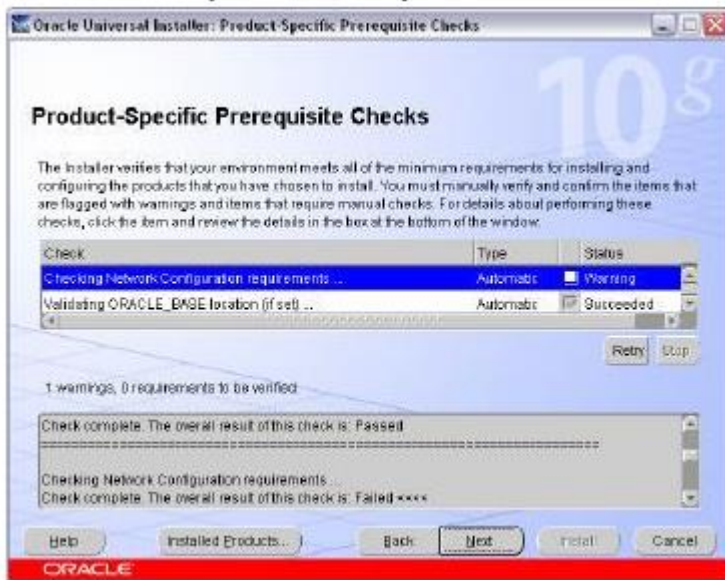
If you have limited space, select **standard edition**. **Personal edition** installs the same software as the Enterprise Edition, but supports only a single-user development and deployment environment.

Create Starter Database — Check this box to create a database during installation. Oracle recommends that you create a starter database for first Create Starter Database — time installations. Choose a **Global Database Name**, like cs157b, or just use the default value.

Type a password. **Don't lose this password**, since you will need it to connect to the database server.

Click next

4. The **Product-Specific Prerequisite Checks** window appears: Click next



5. A summary screen appears showing information such as your global settings, space requirements and the new products to be installed. Click Install to start the installation.

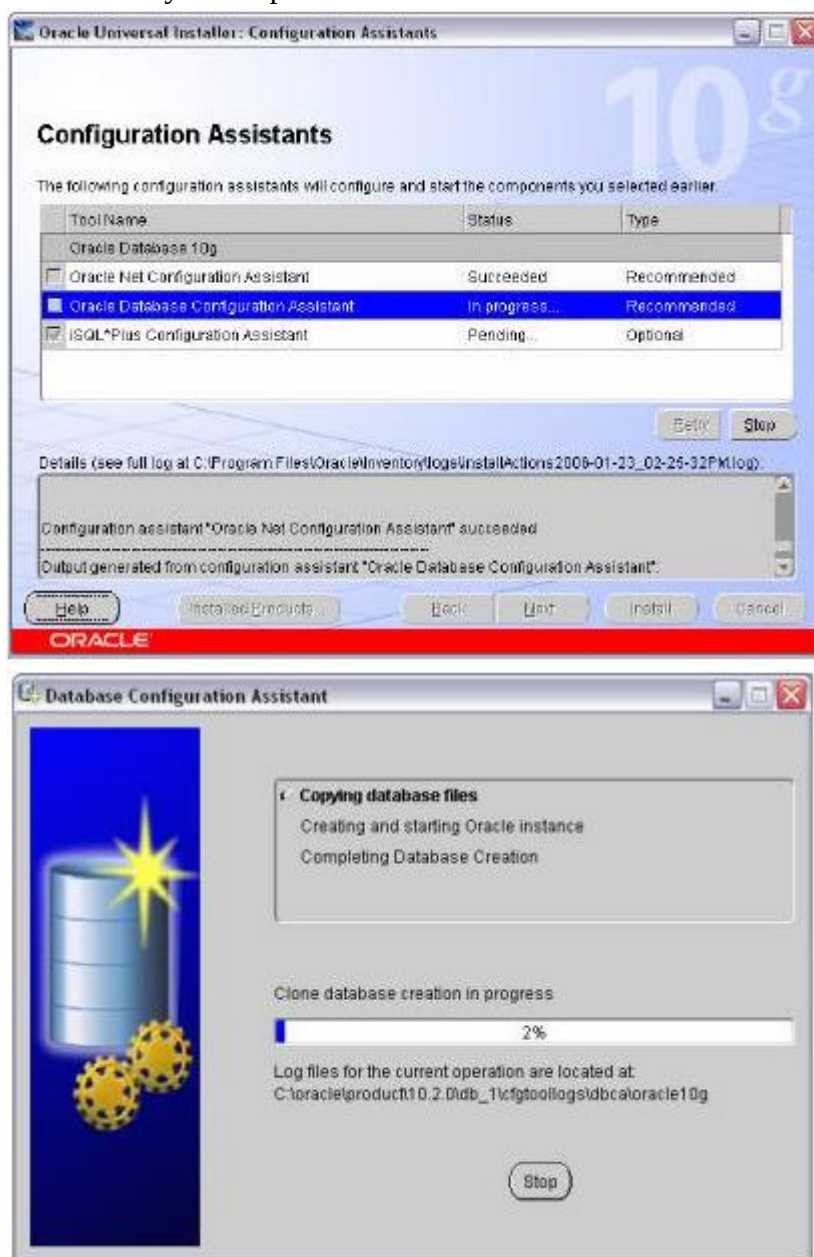


6. The Install window appears showing installation progress.



7. At the end of the installation phase, the **Configuration Assistants** window appears. This window lists the configuration assistants that are started automatically.

If you are creating a database, then the **Database Configuration Assistant** starts automatically in a separate window.

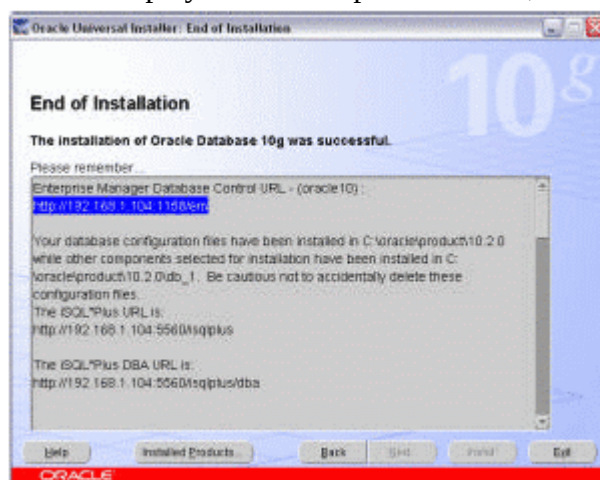


At the end of database creation, you are prompted to unlock user accounts to make the accounts accessible. The **SYS** and **SYSTEM** accounts are already unlocked. **Click OK** to bypass password management.



Note: Oracle 10g still keeps scott / tiger username and password (UID=scott, PWD=tiger) from the old version of oracle. In the old version of oracle, scott/tiger user ID is available by default, but not in oracle 10g. If you want to use scott /tiger account, you must unlock it by clicking “Password Management” at the last window. Password Management window will appear like the one shown below. Find the user name “Scott” and uncheck the “**Lock Account?**” column for the user name.

8. Your installation and database creation is now complete. The End of Installation window displays several important URLs, one of which is for Enterprise Manager.



9. You can navigate to this URL in your browser and log in as the SYS user with the associated password, and connect as SYSDBA. You use Enterprise Manager to perform common database administration task.

Note: you can access Oracle Enterprise Manager using browser by typing the URL shown above in your browser. Instead of typing the IP address, you can also access the Enterprise Manager by typing **<http://localhost:1158/em>** or “**[http://\[yourComputerName\]:1158/em](http://[yourComputerName]:1158/em)**” or by clicking “**Start >> All Programs >> Oracle – [YourOracleHome_home1] >> Database Control – [yourOracleID]**” in

Windowsmenu.

By default, use user ID “**SYSTEM**”, with the password that you have chosen at the beginning of installation, to connect to database, SQLPlus, etc. If you want to use other user ID, you may create a new user.

CONCLUSION:

Thus, we studied about Introduction to DBMS LAB.